Full Incident Response — Incident Report

Intern: UBI-2026-0210

Track: SOC_ANALYSIS

Date: 11/09/2026

1. Case Authority and Scope

This is an incident reconstructed from a sealed, hash-verified synthetic evidence pack (soc-analysis-stage-9-shared-b1.tar.gz, SHA-256: df8fb093a6a69ca885a0f22518f842cf6a470232a6df821fb759a507389f2022). All findings below are derived exclusively from the assigned sealed sources: sysmon.jsonl, powershell-operational.jsonl, initial-access.eml and incident.pcap — processed through the submitted, tested parser and timeline pipeline (timeline-builder/). No conclusion in this report was hand-typed into the timeline; timeline.csv is generated code output, with the exception of the analyst-judgment columns (attack_technique, confidence, primary_or_supporting, alternative_interpretation), which are supplied via timeline-builder/annotations.py and reviewed.

2. Acquisition and Integrity Findings

All four evidence files were verified byte-identical to case-manifest.csv's declared SHA-256 hashes, and the sealed archive itself was verified against the private overlay's archive-level SHA-256 before extraction. An exhaustive search for a manifest-tampered item, duplicate events, malformed timestamps and Zeek-flagged network anomalies (weird.log) found none across all four evidence types. This is reported as a genuine non-finding, not a gap: the case's description of a tampered item was tested for specifically and not reproduced in this variant's acquired evidence.

Check	Method	Result
File-level hash match (x4)	SHA-256 vs case-manifest.csv	All 4 matched
Archive-level hash match	SHA-256 vs private overlay	Matched
Duplicate sysmon lines/event_ids	Exhaustive scan, 100,000 lines	0 found
Malformed timestamps	ISO-8601 parse, all sources	0 found
Zeek weird.log anomalies	Full pcap re-run	None flagged

3. Clock Correction

The sysmon source's clock was found to be offset +85,609 seconds (~23h46m49s) ahead of the true time observed independently by the network capture and the PowerShell operational log. This was confirmed via three independent event-pair measurements using a shared cross-source correlator (evidence_binding token 8d919c5bfb955d01), each measuring a different point in the same causal chain:

Pair	Basis	Measured delta (s)
Primary	curl.exe network_connect (sysmon) vs. conn.log connection start — same discrete network event, two vantage points	85609.0
Corroborating	WINWORD.EXE open (sysmon) vs. PowerShell script-block log	85597.0
Corroborating	sysmon file_delete vs. files.log 8-byte record	85559.95

The three measurements agree within 50 seconds on an ~85,600-second span. The value is not a round number (not 24h, not a whole-hour multiple), consistent with an unsynchronized/drifted host clock rather than a timezone misconfiguration. The 85,609-second offset (curl.exe initiating the connection vs. Zeek's conn.log recording that same connection's start) was selected as the primary measurement because both sides record the identical discrete network event — a TCP connect — from two different vantage points, leaving no real-world elapsed time between them to account for; the other two measured pairs (85,597s and 85,559.95s) span causally separate steps in the chain and were retained only as corroborating evidence, not as the basis for the applied correction. A -85,609s

correction was applied to all sysmon-sourced timestamps. No second clock offset was identified in any other source despite the same searches being applied; this is reported as a limitation, not assumed away.

4. Confirmed Incident Chain

A single, causally-ordered chain was confirmed entirely on host NS-WKS-101, identity northstar\analyst1, correlated across sysmon, PowerShell, and Zeek via the shared evidence_binding token. All times below are corrected UTC.

Time (UTC)	Host/Source	Event	ATT&CK
2026-07-14 (Date header absent)	initial-access.eml	Email delivered to analyst1@northstar.invalid, subject "Case export reconciliation 8d919c5b"	T1566.001
11:59:52	sysmon:50001	WINWORD.EXE opened (parent explorer.exe)	T1204.002
11:59:53	sysmon:50002	powershell.exe spawned by WINWORD.EXE	T1059.001
11:59:56	sysmon:50003	schtasks.exe creates task "Northstar Support Sync"	T1053.005
11:59:57	sysmon:50004	tar.exe creates case-export.zip from synthetic-records.csv	T1560.001
12:00:00	sysmon:50005 / conn.log:9	curl.exe opens outbound connection to 203.0.113.195:8443	T1071.001
12:00:04.004	http.log:9 / files.log:9	HTTP POST /collect/8d919c5b.../case-export.zip (53,891 bytes)	T1041
12:00:01	sysmon:50006	powershell.exe deletes case-export.zip	T1070.004

5. Required Case Questions

What delivered initial access, and what exact artifact executed

Delivery vector confirmed: an email (initial-access.eml) to ***analyst1@northstar.invalid***, correlated to the host execution chain via the shared evidence_binding token. **Limitation:** the email's actual attachment (case-instructions-8d919c5b.txt) is a plain-text file, not the Office document shown executing in sysmon (WINWORD.EXE). The exact bytes of the artifact WINWORD.EXE opened were not recoverable from the sealed evidence — the link between the email and the execution is established by shared correlation token and temporal proximity, not by recovering the literal executed file. This is stated as a confidence-medium finding, not overstated as high.

What C2 infrastructure was used, and when was it active?

Domain sync-v1.updates-example.invalid, IP 203.0.113.195:8443. Active window (corrected UTC): 11:59:56 (connection initiation) through 12:00:50.05 (final files.log record). Confidence: high, confirmed independently via conn.log, http.log, and a raw TCP payload reconstruction that matches the Zeek-extracted archive byte-for-byte.

How did privileges increase, and which identity was affected?

No privilege escalation was observed. Every event in the confirmed chain executes as northstar\analyst1 — no elevation, token manipulation, or identity change appears anywhere in the available evidence. This is reported as a deliberate finding: the absence of escalation evidence, not an unexamined gap.

Which host-to-host movement is confirmed rather than inferred?

None. queries/find_signal_events.py exhaustively scanned all 100,000 sysmon records across all 379 distinct hosts in the file for non-noise event IDs; signal events were found on exactly one host, NS-WKS-101. Impact boundary is confirmed limited to this single host. No lateral movement is claimed or inferred.

What was accessed, what was staged, and what was confirmed exfiltrated?

Staged and exfiltrated: case-export.zip, created by tar.exe from synthetic-records.csv, uploaded via HTTP POST to the confirmed C2 endpoint — confirmed via two independent extraction methods (Zeek file-analysis framework and raw scapy TCP reassembly), which agree byte-for-byte (zip content located at raw-payload offset 229; tail length matches zip size exactly). **Limitation:** the evidence does not show how or when synthetic-records.csv itself was originally created or accessed prior to staging only its inclusion in the archive is confirmed.

What exact synthetic records were recovered from the exfiltration stream?

5,137 data records (5,138 total lines including header) in synthetic-records.csv, columns record_id, customer_ref, case_type, region. Record count derived via structured CSV parsing and independently cross-checked with `wc -l` line count — both agree exactly, confirming no embedded newlines in quoted fields. Archive

SHA-256: 8e994147097a29a84a1b162cc2aaae6f5592ff135406aaca9c1668d512b98825, 53,891 bytes, 2 files (synthetic-records.csv + evidence-marker.txt). The archive's evidence-marker.txt reads "UBI-A9-SHARED-B1" — the shared base-pack marker.

What evidence would most strongly challenge the primary hypothesis?

If "Northstar Support Sync" were shown to match a real, IT-documented scheduled-sync tool naming convention in this organization's environment, that would directly undercut the persistence-technique classification (T1053.005) and support the benign-administrative-sequence alternative instead. No such documentation exists in the sealed evidence, so this remains the leading hypothesis, but it is the single piece of evidence most capable of overturning it.

6. Detections and Response

One Sigma rule (Office-to-script-to-archive-to-exfil process chain) and one Zeek notice rule (confirmed C2 domain) were written and tested against positive and negative fixtures. The Sigma rule's positive fixture uses the confirmed real chain shape; its negative fixture uses a benign explorer-spawned PowerShell sequence. The Zeek rule's positive case is the real sealed pcap; its negative case is a self-constructed synthetic benign-HTTP pcap, since no comparison benign capture evidence supplied — noted as a limitation.

Containment (dry-run-by-default, require explicit --apply/-Confirm to execute): network block of the confirmed C2 IP and a DNS sinkhole entry for the confirmed C2 domain; removal of the single confirmed persistence mechanism ("Northstar Support Sync" scheduled task). Recovery checklist scoped explicitly to the single confirmed host (NS-WKS-101), recommending re-image over manual cleanup given the initial-access vector was a macro-capable Office document, and flagging credential rotation as precautionary rather than evidence-driven, since no credential theft was observed.

7. Known Limitations

a. The literal executed initial-access artifact (the Office document itself) was not recovered; delivery is inferred from correlation token and timing, not from the file's bytes.

b. A second clock offset (implied by the case description) was not found in any source despite exhaustive checks.

c. No DNS traffic exists in incident.pcap; the zeek_dns parser is verified only against the public synthetic fixture.

d. No real .evtx evidence exists in this pack; PARSER-EVTX is verified only against the public synthetic fixture.

e. Zeek's uid field is non-deterministic across reruns; downstream logic re-derives it live rather than hardcoding it, but this means raw Zeek log outputs are not byte-identical across rebuilds (the final timeline.csv's event ordering and content are unaffected, since correlation relies on evidence_binding, not uid).

f. remove-persistence.ps1 was verified via static content checks only; no Windows/pwsh runtime was available to execute it live in this environment.